

CSE 312 – Continuous Probability

Lectures 14, 15 & 16 | Winter 2026

Lecture 14: Introduction to Continuous Probability

Why Continuous Random Variables?

So far all sample spaces have been discrete (countable). We now need tools for uncountably infinite sample spaces.

- Examples: time until a bus arrives, where a dart lands on a board.
- Even for large discrete spaces, continuous models are often mathematically easier — model as continuous, then round.
- **Key issue:** We cannot assign nonzero probability to individual outcomes. Probabilities are defined over intervals.

Probability Density Function (PDF)

The PDF, written $f_X(k)$, is the continuous analogue of the PMF. Key properties:

- $f_X(k) \geq 0$ for all k
- $\int f_X(k) dk = 1$ (integrating over all reals)
- **Probabilities via integration:** $P(a \leq X \leq b) = \int \text{from } a \text{ to } b \text{ of } f_X(z) dz$

Note: $f_X(x)$ is NOT a probability — it is a density. $P(X = \text{any single value}) = 0$ for continuous RVs.

The ratio $f_X(a)/f_X(b)$ gives the relative likelihood of X being near a vs near b .

Impossible vs. Probability-Zero Events

- **Discrete:** Probability 0 means impossible.
- **Continuous:** All impossible events have prob 0, but NOT all prob-0 events are impossible. E.g., $P(X = 0.1) = 0$ yet $X = 0.1$ can still occur.

Cumulative Distribution Function (CDF)

CDF: $F_X(k) = P(X \leq k) = \text{integral from } -\infty \text{ to } k \text{ of } f_X(z) dz$

- **PDF from CDF:** Differentiate — $f_X(k) = d/dk F_X(k)$
- For continuous RVs, $<$ and $\leq$ are interchangeable (single points have probability 0).

Expectation and Variance

$E[X]$: integral of $z * f_X(z) dz$ over all reals

$E[g(X)]$: integral of $g(z) * f_X(z) dz$ (Law of the Unconscious Statistician)

$\text{Var}(X)$: $E[X^2] - (E[X])^2$

- Linearity of Expectation still holds: $E[aX + bY + c] = a \cdot E[X] + b \cdot E[Y] + c$

Uniform Distribution: Unif(a, b)

$X \sim \text{Unif}(a, b)$ is a uniformly random real number between a and b .

PDF: $f_X(k) = 1/(b-a)$ for $a \leq k \leq b$, else 0

CDF: $F_X(k) = 0$ if $k < a$

E[X]: $(a + b) / 2$

Var(X): $(b - a)^2 / 12$

Lecture 15: The Continuous Zoo

A summary of the three core continuous distributions covered in this course.

Unif(a,b): $f = 1/(b-a)$, $E[X] = (a+b)/2$, $\text{Var} = (b-a)^2/12$

Exp(lambda): $f = \lambda e^{-\lambda k}$ for $k \geq 0$, $E[X] = 1/\lambda$, $\text{Var} = 1/\lambda^2$

N(mu, sigma^2): $f = (1/\sigma \sqrt{2\pi}) * \exp(-(x-\mu)^2/(2\sigma^2))$,
 $E[X] = \mu$, $\text{Var} = \sigma^2$

Lecture 16: Exponential and Normal Distributions

Exponential Distribution: Exp(λ)

$X \sim \text{Exp}(\lambda)$ models the waiting time until the next event. λ is the average number of events per unit time.

PDF: $f_X(k) = \lambda * e^{-\lambda k}$ for $k \geq 0$; else 0

CDF: $F_X(k) = 1 - e^{-\lambda k}$ for $k \geq 0$; else 0

E[X]: $1 / \lambda$

Var(X): $1 / \lambda^2$

Intuition: Connection to Poisson

- **Poisson asks:** How many events occur in a fixed interval?
- **Exponential asks:** How long until the next event?
- **Deriving the CDF:** Let $X \sim \text{Poi}(\lambda t)$. Then $P(Y > t) = P(X = 0) = e^{-\lambda t}$, so $F_Y(t) = 1 - e^{-\lambda t}$.
- **Getting the PDF:** Differentiate the CDF: $f_Y(t) = \lambda * e^{-\lambda t}$
- Higher λ = steeper drop = events happen sooner on average.

Memorylessness

The exponential is the continuous analogue of the geometric distribution — it has no memory.

Memoryless property: $P(X \geq s+t \mid X \geq s) = P(X \geq t)$

- Intuitively: having already waited s units of time tells you nothing about when the next event occurs.
- **Continuous note:** For continuous RVs, $P(X < a) = P(X \leq a)$, so $<$ and $\leq$ are freely interchangeable (unlike discrete RVs).

Normal (Gaussian) Distribution: $N(\mu, \sigma^2)$

One of the most important distributions in all of probability and statistics. Real-world data very often follows a bell curve shape.

PDF: $f_X(x) = (1 / (\sigma * \sqrt{2\pi})) * \exp(-(x-\mu)^2 / (2\sigma^2))$

CDF: No closed form – use the standard normal table (Φ)

E[X]: μ

Var(X): σ^2

- μ : Controls the center. Shifting μ shifts the entire curve left or right.
- σ^2 : Controls the spread. Larger σ^2 = flatter/wider curve. Smaller σ^2 = taller/narrower.
- Symmetric around μ , which is why $E[X] = \mu$.

Scaling Normals

Linear transformations of normals produce another normal — a special and very useful property.

If $X \sim N(\mu, \sigma^2)$ and $Y = aX + b$: then $Y \sim N(a\mu + b, a^2 * \sigma^2)$

- This does NOT hold for most other distributions (e.g., multiplying a Binomial by 3/2 no longer gives a Binomial).

Standardizing (Z-scores)

To convert $X \sim N(\mu, \sigma^2)$ to a standard normal $Z \sim N(0,1)$:

Standardize: $Z = (X - \mu) / \sigma$

- This lets us use a single Φ table for all normal distribution calculations.
- **Procedure:** (1) Standardize via $Z = (X-\mu)/\sigma$, (2) round z to hundredths, (3) look up $\Phi(z)$ in table.
- **Key identity:** $\Phi(-z) = 1 - \Phi(z)$ (use this for negative z-scores)

Worked Examples

Example 1: $Y \sim N(5, 4)$, find $P(Y > 9)$

- $Z = (9-5)/2 = 2.00$
- $P(Y > 9) = 1 - \Phi(2.00) = 1 - 0.97725 = 0.02275$

Example 2: $X \sim N(3, 2)$, find $P(1 \leq X \leq 4)$

- Lower z : $(1-3)/\sqrt{2} \approx -1.41$
- Upper z : $(4-3)/\sqrt{2} \approx 0.71$
- $P = \Phi(0.71) - \Phi(-1.41) = 0.76115 - (1 - 0.92073) \approx 0.682$

The 68-95-99.7 Rule

- $P(|X - \mu| \leq 1\sigma) \approx 0.68$
- $P(|X - \mu| \leq 2\sigma) \approx 0.9545$
- $P(|X - \mu| \leq 3\sigma) \approx 0.997$

Scaling vs. Summing IID Copies

Note: Common confusion: $2X$ and $X1+X2$ have the same expectation but DIFFERENT variances!

- **2X:** Roll ONE die, double the result. Support = {2,4,6,8,10,12}. Cannot get 7.
- **X1 + X2:** Roll TWO independent dice and add. Support = {2,3,...,12}. All values reachable.

Expectations are equal: $E[2X] = E[X1 + X2] = 2 * E[X]$

Variances differ: $Var(2X) = 4 * Var(X)$ but $Var(X1+X2) = 2 * Var(X)$

Quick-Reference Summary

Distribution	PDF $f_X(k)$	CDF $F_X(k)$	$E[X]$	$Var(X)$
Unif(a, b)	$1/(b-a)$ for $a \leq k \leq b$	$(k-a)/(b-a)$	$(a+b)/2$	$(b-a)^2/12$
Exp(lambda)	$\lambda * e^{(-\lambda k)}$ $k \geq 0$	$1 - e^{(-\lambda k)}$	$1/\lambda$	$1/\lambda^2$
$N(\mu, \sigma^2)$	$(1/\sigma * \sqrt{2\pi}) * e^{(...)}$	Use Phi table	μ	σ^2